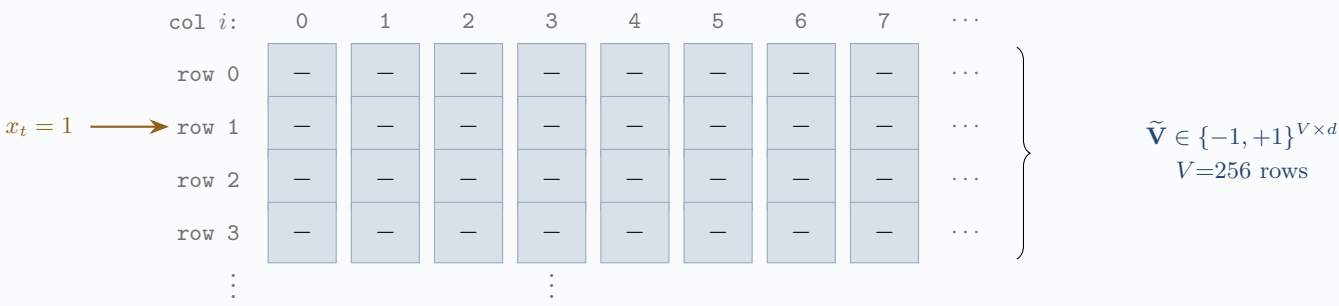


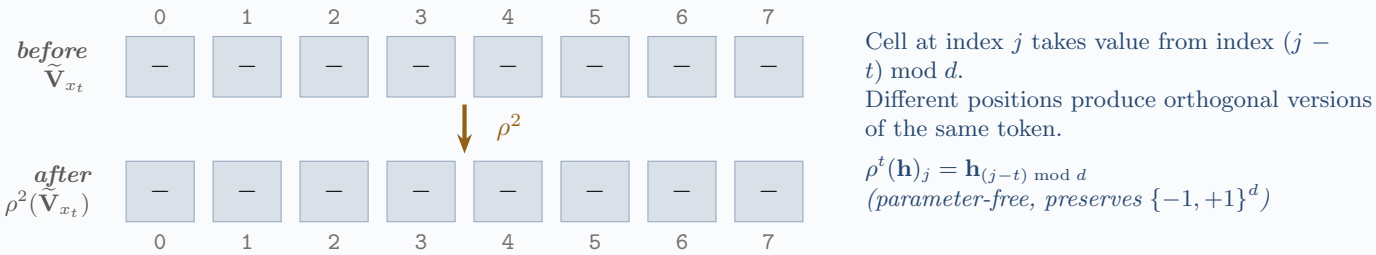
HDC-RWKV (nb12c) — Full Architecture Expansion

$V=256, d=256, L=1$, block size 16 • 16,434 deployment bytes • every learnable parameter bipolar at inference

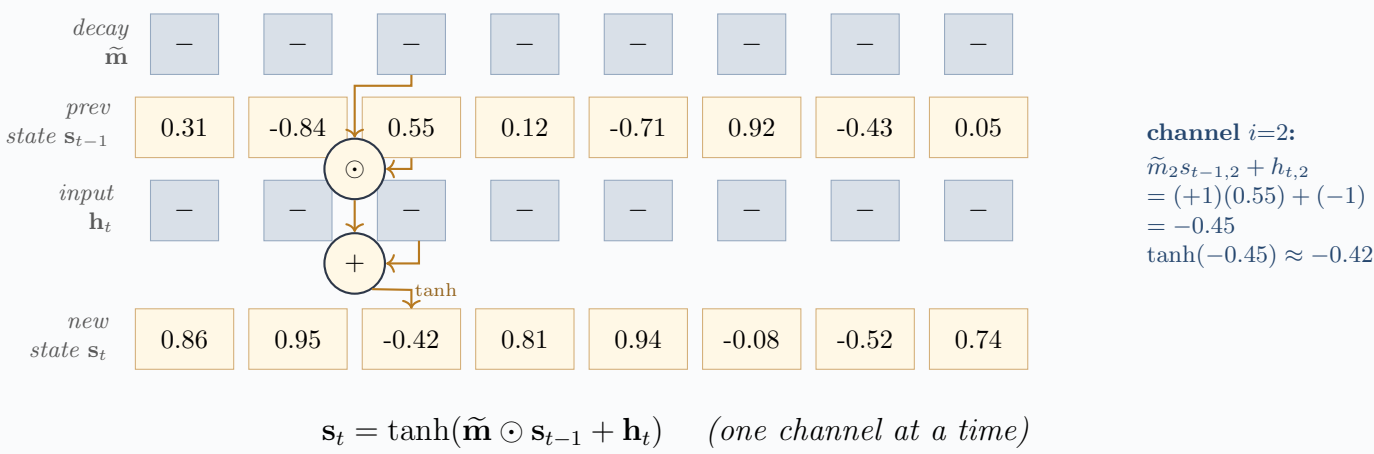
Block 1. Token lookup — bipolar embedding table



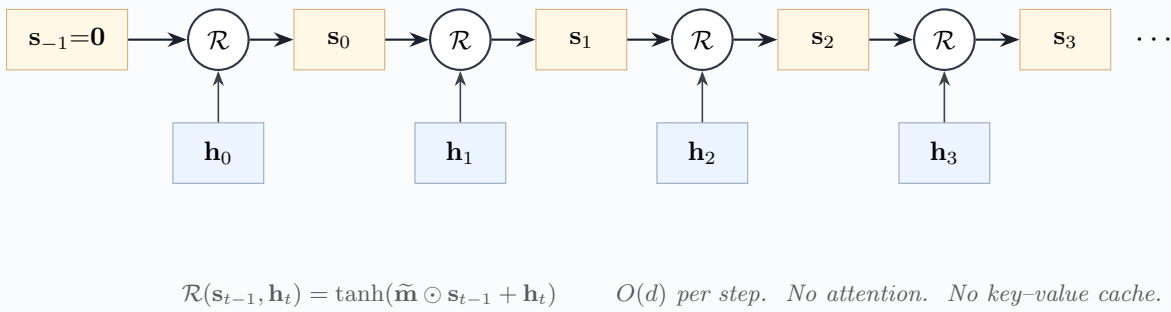
Block 2. Position binding ρ^t (cyclic right-rotation)



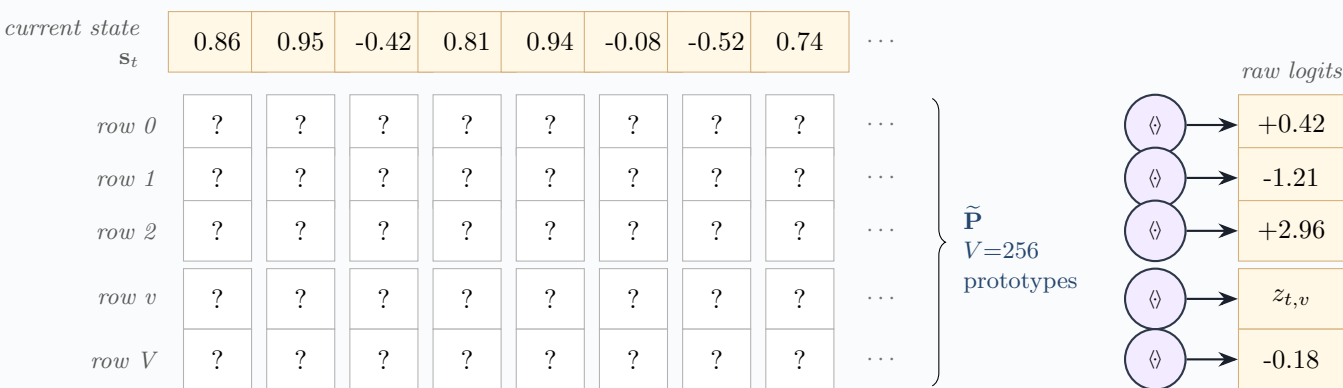
Block 3a. Recurrence — per-channel detail at one timestep t



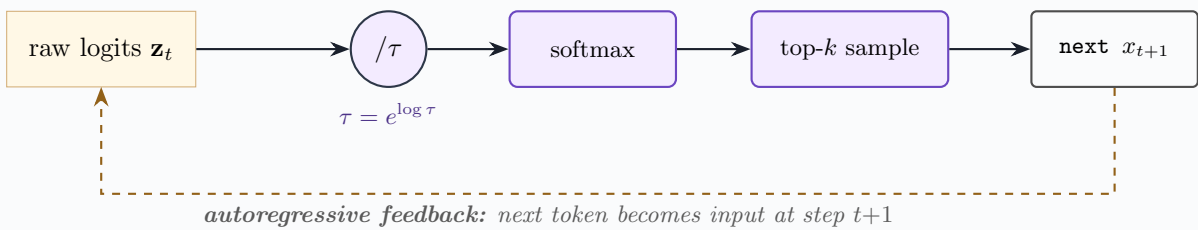
Block 3b. Recurrence — unrolled across time



Block 4a. Prototype-similarity output — one dot product per vocabulary item



Block 4b. Temperature scaling, softmax, top- k sampling, autoregressive feedback

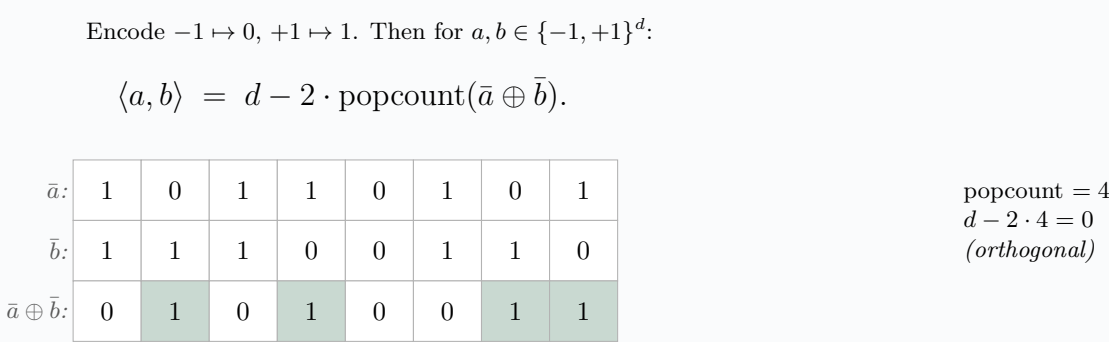


Aux A. Straight-Through Estimator ($\widetilde{\text{sign}}$)



Forward is hard binarisation; backward is identity on the active strip $[-1, 1]$ and zero outside. The continuous shadow $\tilde{\mathbf{V}}$ is updated by backpropagation; the recurrence always sees $\tilde{\mathbf{V}} \in \{-1, +1\}^{V \times d}$.

Aux B. Deployment equivalent — XOR + popcount



The 6502 inner loop: XOR two $d/8$ -byte rows, popcount the result, subtract $2 \cdot \text{popcount}$ from d . No multiplications, no floating point, no NaN reachable. For $d=256$: 32 bytes XORed and one popcount lookup per byte.